

Iron-Catalyzed Aerobic Oxidation of Aldehydes: Single Component Catalyst and Mechanistic Studies

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ABSTRACT An aerobic oxidation of aldehydes towards carboxylic acids in MeCN using 1 atm of pure oxygen or oxygen in air as the oxidant and a catalytic amount of single component catalyst, $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$, has been developed. Carboxylic acids with different synthetically useful functional groups were obtained at room temperature. Two mechanistic pathways have been proposed based on isotopic labeling, NMR monitoring, and control experiments. The practicality of this reaction has been demonstrated by conducting several 50 mmol-scale reactions using pure oxygen or an air-flow of ~30 mL/min.

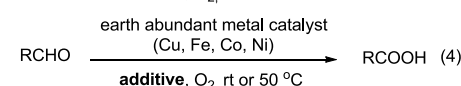
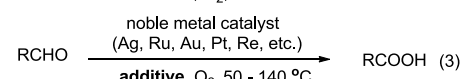
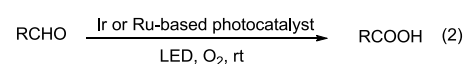
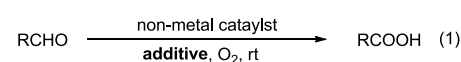
KEYWORDS iron catalysis, aerobic oxidation, aldehyde, carboxylic acid, single component catalyst, mechanistic studies

Introduction

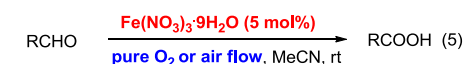
As one of the most important and fundamental chemicals in industry and laboratory, carboxylic acids could be obtained by oxidation reaction of corresponding alcohols or aldehydes.^[1] It is well known that some aldehydes may be oxidized to acids very slowly when exposed to air.^[2] However, the process of autooxidation was too slow to meet the requirement of industrial manufacture or preparation in academic laboratory. Traditionally, strong and stoichiometric oxidants, such as KMnO_4 ,^[3] CrO_3 ,^[4] and hypervalent iodine compounds,^[5] have to be used to convert aldehydes to carboxylic acids. However, the cost and waste are problematic. As molecular oxygen is a clean and atom economic oxidant, aerobic oxidations of aldehydes to carboxylic acids have been pursued by chemists around the world. Non-metal catalysts were also reported (Scheme 1, eqn (1)).^[6] excessive amount of a base was combined with a catalytic amount of NHC to catalyze aerobic oxidation of aldehydes.^[6a–6c] In 2016, a three-component catalyst–ketoABNO, NaNO_2 , and HNO_3 —was applied to catalyze the aerobic oxidation of aldehydes.^[6d] Noble metal photocatalysts or other photosensitive reagents have been used for this transformation with visible light and molecular oxygen^[7] (Scheme 1, eqn (2)). Another protocol is utilizing noble metal catalysts, such as Ag, Ru, Au, Pt, Re complexes,^[8] usually using stoichiometric additives with heating (Scheme 1, eqn (3)). Earth abundant metal catalysts such as Cu, Fe, Co, Ni complexes^[9] have also been reported to catalyze this reaction (Scheme 1, eqn (4)). In 2016, Li's group^[9a] reported a catalytic protocol using 5 mol% of $[\text{Cu}(\text{acac})_2]/\text{SIMes}$ with 1 equiv. of NaOH to achieve the aerobic oxidation of aliphatic aldehydes and aromatic aldehydes; the aerobic oxidation of α,β -unsaturated aldehydes was reported by Sato in 2016^[9b] using two-component catalyst–1 mol% of $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ and 20 mol% of CF_3COONa —in EtOAc; in 2016, we utilized three-component catalyst– $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$, TEMPO, and KCl—for the aerobic oxidation of aldehydes to carboxylic acids.^[10] Recently, Han and Wei^[9c] reported two-component—the Fe/Mo-based heterogeneous catalyst and 10 mol% of Na_2CO_3 —to catalyze the oxidation reaction at 50 °C. Thus, development of efficient catalytic simple aerobic oxidation of aldehydes utilizing a cheap catalyst such as an iron catalyst^[11] and an atmospheric pressure of oxygen or air at room temperature is still of high interest. In this report, we wish to report our recent observation that $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ may act as the single-component catalyst to accomplish aerobic oxidation reaction of aldehydes to carboxylic acids even at room temperature (Scheme 1, eqn (5)).

Scheme 1 The catalytic aerobic oxidations of aldehydes to carboxylic acids

Known protocols:



This work:



Results and Discussion

In the control experiment of oxidation of dodecanal **1a**,^[10] we observed that with 10 mol% each of $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ and TEMPO in DCE at room temperature, 81% NMR yield of carboxylic acid **2a** was formed together with 14% NMR recovery of aldehyde **1a** (Table 1, entry 2). However, upon using 5 mol% of $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ alone, **1a** was converted into 64% NMR yield of carboxylic acid **2a** together with 32% NMR yield of carboxylic peracid **3a**^[10] (Table 1, entry 3). By just adding 1 mol% of TEMPO, the formation of carboxylic peracid **3a** was suppressed with 26% NMR yield of **2a** and 74% NMR recovery of **1a** indicating that a radical process may be involved during the reaction (Table 1, entry 4). The selectivity of **2a/3a** was much better in anhydrous DCE or DCM (Table 1, entries 5 and 6). Surprisingly, in anhydrous MeCN, 94% NMR yield of **2a** was obtained exclusively (Table 1, entry 7), which led to the discovery of the single-component iron catalyst for the aerobic oxidation of aldehydes for the first time.

We then evaluated the effect of metal nitrates on the aerobic oxidation reaction of aldehyde **1a**. KNO_3 and $\text{Mn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ gave 85% and 98% NMR recovery of **1a**, respectively (Table 2, entries 1 and 2). $\text{Ce}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ gave 30% NMR yield of **2a** and 60% NMR recovery of **1a** (Table 2, entry 3). With $\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$, 78% NMR yield of **2a**, 18% NMR yield of peracid **3a**, and 4% NMR recovery of **1a** were observed (Table 2, entry 4). When $\text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ and

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Table 1 Optimization of the aerobic oxidation reaction of aldehyde **1a**^a

$n\text{-C}_{11}\text{H}_{23}\text{CHO} \xrightarrow[\text{TEMPO, O}_2, \text{solvent, 25 } ^\circ\text{C, 12 h}]{\text{Fe(NO}_3)_3 \cdot 9\text{H}_2\text{O (5 mol\%)}} n\text{-C}_{11}\text{H}_{23}\text{COOH} + n\text{-C}_{11}\text{H}_{23}\text{CO}_3\text{H}$						
Entry	TEMPO (mol%)	KCl (mol%)	Solvent ^b	Yield of 2a ^c /%	Yield of 3a ^c /%	Recovery of 1a ^c /%
1 ^d	10	10	DCE	95	—	—
2 ^d	10	0	DCE	81	—	14
3	0	0	DCE	64	32	—
4	1	0	DCE	26	—	74
5	0	0	DCE ^e	72	28	—
6	0	0	DCM	84	~7	—
7	0	0	MeCN ^e	94	—	—

^aThe reaction was carried out on a 1.0 mmol scale of **1a** in 4.0 mL of solvent at 25 °C with a bag of O₂. ^bThe solvents were directly used from commercial suppliers, if not specifically mentioned. ^cNMR yield. ^d10 mol% of Fe(NO₃)₃·9H₂O was added. ^eAnhydrous solvent was added.

Ni(NO₃)₂·6H₂O were used, aldehyde **1a** could be fully converted with generation of 10% and 16% NMR yields of peracid **3a** in addition to the carboxylic acid **2a** (Table 2, entries 5 and 6). FeCl₃·6H₂O gave 52% NMR yield of **2a**, 15% NMR yield of **3a**, and 3% NMR recovery of **1a** (Table 2, entry 7). With Fe₂(SO₄)₃, only 15% NMR yield of **2a** was generated with 7% NMR yield of **3a** and 55% NMR recovery of aldehyde **1a** (Table 2, entry 8). Thus, we defined 5 mol% of Fe(NO₃)₃·9H₂O in anhydrous MeCN using a bag of molecular oxygen at 25 °C as the standard conditions for further studies (Table 1, entry 7).

Table 2 Effect of metal salts on the aerobic oxidation reaction of aldehyde **1a**^a

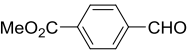
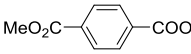
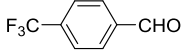
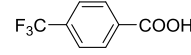
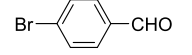
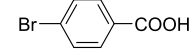
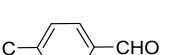
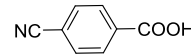
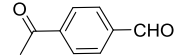
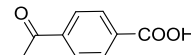
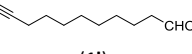
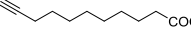
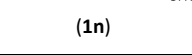
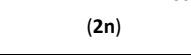
$n\text{-C}_{11}\text{H}_{23}\text{CHO} \xrightarrow[\text{O}_2, \text{anhydrous MeCN, 25 } ^\circ\text{C, 12 h}]{\text{Metal salt (5 mol\%)}} n\text{-C}_{11}\text{H}_{23}\text{COOH} + n\text{-C}_{11}\text{H}_{23}\text{CO}_3\text{H}$				
Entry	Metal salt	Yield of 2a ^b /%	Yield of 3a ^b /%	Recovery of 1a ^b /%
1	KNO ₃	14	—	85
2	Mn(NO ₃) ₂ ·6H ₂ O	—	—	98
3	Ce(NO ₃) ₂ ·6H ₂ O	30	—	60
4	Cu(NO ₃) ₂ ·3H ₂ O	78	18	4
5	Co(NO ₃) ₂ ·6H ₂ O	81	10	—
6	Ni(NO ₃) ₂ ·6H ₂ O	83	16	—
7	FeCl ₃ ·6H ₂ O	52	15	3
8	Fe ₂ (SO ₄) ₃	15	7	55

^aThe reaction was carried out on a 1.0 mmol scale of **1** in 4.0 mL of anhydrous MeCN at 25 °C with a bag of O₂. ^bNMR yield.

With optimized conditions in hand, we investigated the substrate scope of the aerobic oxidation. Apart from linear aliphatic aldehydes (Table 3, entries 1 and 2), cyclohexyl formaldehyde **1c** and 3-phenylpropanal **1d** could be oxidized smoothly to corresponding acids (Table 3, entries 3 and 4). A variety of synthetic useful functional groups such as ester, trifluoromethyl, bromo, cyano, and carbonyl groups were well tolerated in this reaction (Table 3, entries 5—11). Synthetically attractive alkynes are much less studied in the literature.^[10,13] Here, terminal alkynal **1l**, TMS substituted alkynal **1m**, and methyl substituted alkynal **1n** were also oxidized efficiently to acids in decent yields (Table 3, entries 12—14).

The reaction may be easily conducted on 50 mmol scale by using a pure oxygen bag or a slow air flow (30 mL/min) (Scheme 2).

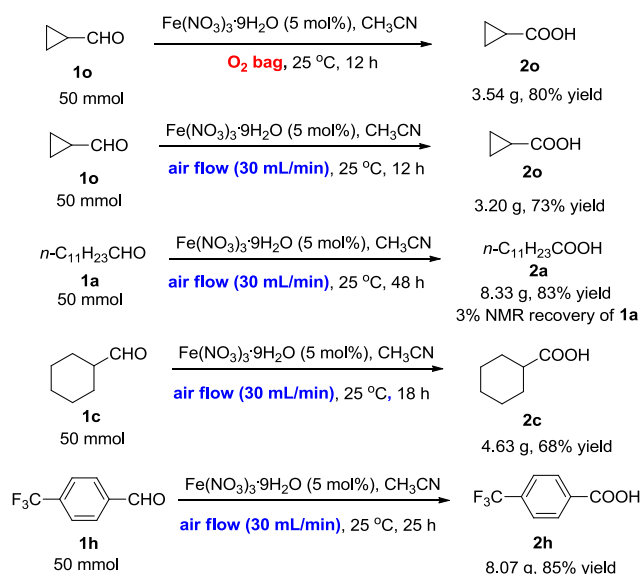
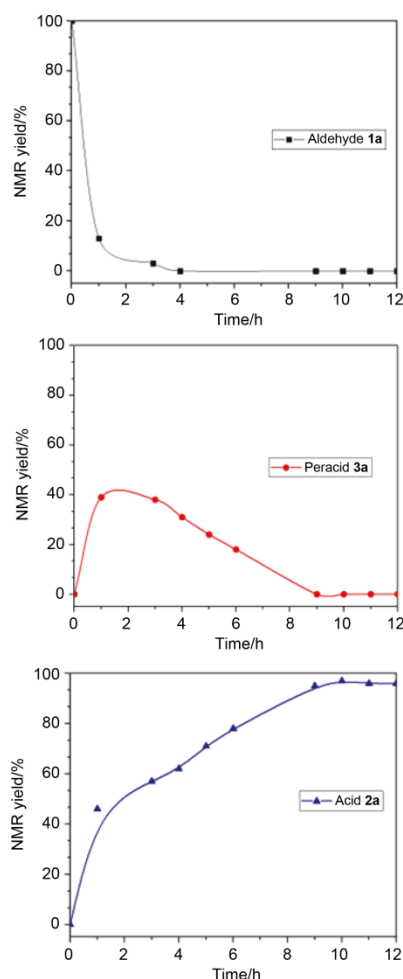
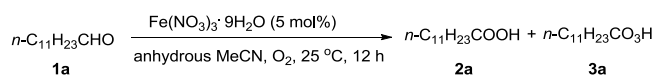
Table 3 The substrate scope of aerobic oxidation of aldehydes^a

$\text{RCHO} \xrightarrow[\text{O}_2, \text{anhydrous MeCN, 25 } ^\circ\text{C, } t \text{ (h)}]{\text{Fe(NO}_3)_3 \cdot 9\text{H}_2\text{O (5 mol\%)}} \text{RCOOH}$				
Entry	Substrate	Product 2	t/h	Yield ^b /%
1	<i>n</i> -C ₁₁ H ₂₃ CHO (1a)	<i>n</i> -C ₁₁ H ₂₃ COOH (2a)	12	82
2	<i>n</i> -C ₁₅ H ₃₁ CHO (1b)	<i>n</i> -C ₁₅ H ₃₁ COOH (2b)	36	76 ^c
3	CyCHO (1c)	CyCOOH (2c)	24	52
4	Ph(CH ₂) ₂ CHO (1d)	Ph(CH ₂) ₂ COOH (2d)	48	63
5	AcO(CH ₂) ₇ CHO (1e)	AcO(CH ₂) ₇ COOH (2e)	12	60
6	MeO ₂ C(CH ₂) ₄ CHO (1f)	MeO ₂ C(CH ₂) ₄ COOH (2f)	12	65
7			24	80
	(1g)	(2g)		
8			12	79
	(1h)	(2h)		
9			18	64
	(1i)	(2i)		
10			48	72 ^d
	(1j)	(2j)		
11			48	68 ^e
	(1k)	(2k)		
12			24	67
	(1l)	(2l)		
13			24	63
	(1m)	(2m)		
14			48	75
	(1n)	(2n)		

^aThe reaction was carried out on a 1.0 mmol scale of **1** in 4.0 mL of anhydrous MeCN at 25 °C with a bag of O₂. ^bIsolated yield. ^c3% NMR recovery of the aldehyde. ^d12% recovery of the aldehyde. ^e16% NMR recovery of the aldehyde.

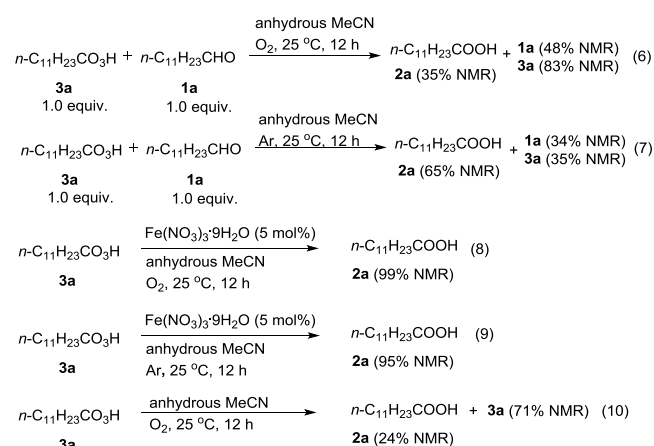
In order to unveil the mechanism, the aerobic oxidation reaction of aldehyde **1a** was monitored. We observed that aldehyde **1a** was consumed within the first 4 h, with the generation of acid **2a** and peracid **3a**. Then the amount of peracid **3a** decreased. Within 12 h, all peracid **3a** disappeared with 96% NMR yield of carboxylic acid **2a** as the final product (Scheme 3).

Some control experiments were then conducted (Scheme 4): At first, 1.0 equiv. of peracid **3a** and 1.0 equiv. of aldehyde **1a** were mixed in anhydrous MeCN in O₂ condition. After 12 h, 35% NMR yield of acid **2a** was generated with 48% NMR recovery of **1a** and 83% NMR recovery of **3a** (eqn (6)); when the same mixture

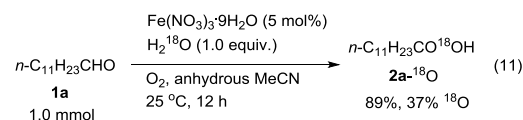
Scheme 2 Large-scale reactions with O₂ or slow air flow**Scheme 3** Reaction profiles of aerobic oxidation reaction of aldehyde **1a**

was stirred under Ar atmosphere for 12 h, 65% NMR yield of acid **2a** was obtained with 34% NMR recovery of **1a** and 35% NMR recovery of **3a** (eqn (7)); when peracid **3a** alone was stirred with

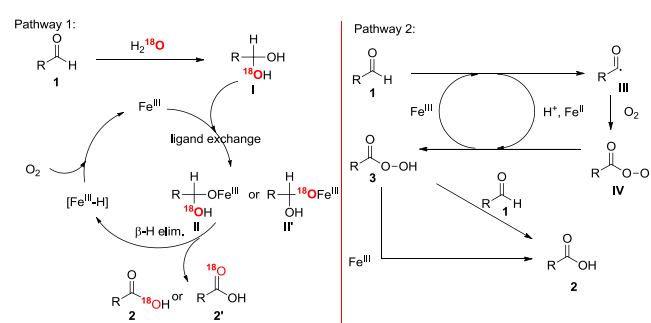
$\text{Fe(NO}_3)_3 \cdot 9\text{H}_2\text{O}$ under O₂ or Ar, almost quantitative yield of carboxylic acid **2a** was afforded (eqn (8) and (9)); even when peracid **3a** was stirred without catalyst for 21 h, only 24% NMR yield of acid **2a** and 71% NMR recovery were observed (eqn (10)).^[12] These control experiments indicated that carboxylic peracid **3a** and aldehyde **1a** could react smoothly to form carboxylic acid **2a** both under both O₂ and Ar atmosphere. In addition, carboxylic peracid **3a** could be efficiently transformed into carboxylic acid **2a** in MeCN catalyzed by $\text{Fe(NO}_3)_3 \cdot 9\text{H}_2\text{O}$.

Scheme 4 Control experiments of **3a** and **1a**

Furthermore, isotopic ¹⁸O distribution experiment was conducted by adding 1.0 equiv. of H₂¹⁸O to the reaction under the standard conditions, and 89% yield of acid **2a**-¹⁸O with 37% ¹⁸O incorporation was isolated (eqn (11)). It illustrated that H₂O also participated in the reaction process.^[10] In addition, the formation of NO and NO₂ was not detected, indicating a different mechanism as compared to that reported in ref. 10.



Thus, two plausible pathways are proposed (Scheme 5): Aldehyde **1** is attacked by H₂¹⁸O to form aldehyde monohydrate **I**. After ligand exchange and β-H elimination, carboxylic acid **2** or **2'** is obtained. The resulting [Fe^{III}-H] is in equilibrium with H⁺ and Fe^{II}, which would be oxidized to Fe^{III} by O₂ to finish the catalytic cycle. The second one involves a radical process: acyl radical **III**, which is formed by oxidation of aldehyde with Fe^{II}, reacts with O₂ to afford peracid radical **IV**. Peracid **3** is generated when **IV** was reduced by Fe^{II} in the presence of H⁺. Carboxylic acid **2** is formed via the reaction between aldehyde **1** and peracid **3** or Fe^{III}-catalyzed highly selective conversion of peracid **3** to carboxylic acid **2**.

Scheme 5 Plausible mechanistic pathways

Conclusions

In summary, an efficient catalytic room temperature aerobic oxidation has been developed. With 5 mol% of $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ as the single catalyst and 1 atm of pure O_2 or oxygen in air as the oxidant, different types of aldehydes with various useful functional groups could be oxidized to acid smoothly. The reaction could be conducted easily on 50 mmol scale with O_2 or a slow air flow. Based on careful mechanistic study, it is believed that the reaction proceeds via two plausible pathways. Further studies including mechanism details are being conducted actively in this laboratory.

Experimental

General information. $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ was purchased from Energy Chemical. Liquid aldehydes were distilled before use. MeCN was refluxed over CaH_2 and distilled right before use. Petroleum ether (60–90 °C) was used for chromatography. Other reagents were used as received without further treatment. Substrates **1b**, **1e**, **1f**, **1l**, **1m**, and **1n** were synthesized via iron-catalyzed aerobic oxidation of corresponding alcohols.^[13]

Caution: Oxygen in use in combination with organic solvents; remove all ignition sources including sources of sparks, static, or flames since oxygen increases intensity of any fire. Inhalation of pure oxygen should be avoided as well.

Synthesis of dodecanoic acid (2a) (Table 3, entry 1): to a Schlenk tube were added $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (20.1 mg, 0.05 mmol), **1a** (184.1 mg, 1.0 mmol), and MeCN (4 mL) sequentially under the atmosphere of oxygen from a gas bag (commercial size: 2 L, could be expended to 5 L). The Schlenk tube was then stirred at 25 °C until completion of the reaction as monitored by TLC (petroleum ether/ethyl acetate = 5/1) (12 h). The crude reaction mixture was filtrated through a short column of silica gel (height: 2 cm, Φ : 3 cm) eluted with diethyl ether (25 $\times$ 3 mL). After evaporation, the residue was purified by chromatography on silica gel to afford **2a** (164.4 mg, 82%) (eluent: petroleum ether/ethyl acetate = 5/1) as a solid: m.p. 43.1–44.0 °C (petroleum ether/ethyl acetate) (reported:^[14] 43–44 °C); ^1H NMR (400 MHz, CDCl_3) δ : 11.79 (brs, 1H, COOH), 2.35 (t, J = 7.6 Hz, 2H, CH_2), 1.69–1.56 (m, 2H, CH_2), 1.40–1.18 (m, 16H, $8 \times \text{CH}_2$), 0.88 (t, J = 6.2 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, CDCl_3) δ : 180.7, 34.1, 31.9, 29.6, 29.4, 29.3, 29.2, 29.0, 24.6, 22.7, 14.1; IR (neat) ν : 3400–2300, 1695, 1468, 1429, 1410, 1301, 1277, 1248, 1219, 1193 cm^{-1} ; MS (EI, 70 eV) m/z (%): 200 (M^+ , 99.62), 73 (100).

The large scale (50 mmol) reaction for synthesis of dodecanoic acid (2a) (Scheme 2): To a 1000 mL three-neck flask were added $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (1.0104 g, 2.5 mmol), **1a** (9.2169 g, 50.0 mmol), and MeCN (150.0 mL) sequentially (Figure 1). The result-

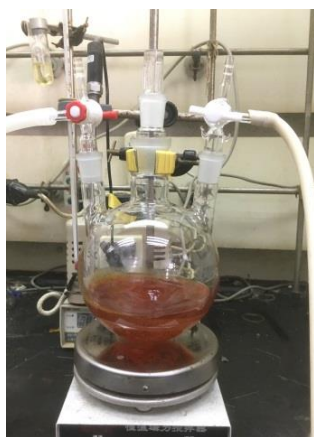


Figure 1 Apparatus for the aerobic reaction with a slow air flow (30 mL/min).

ing mixture was stirred at 25 °C with a slow flow of air from an air cylinder (99.99%, 30 mL/min) passing over the surface of the resulting mixture until completion of the reaction as monitored by TLC (petroleum ether/ethyl acetate = 5/1) (48 h). The crude reaction mixture was filtrated through a short column of silica gel (height: 4.5 cm, Φ : 5 cm) eluted with ethyl ether (4 $\times$ 150 mL). After evaporation, the residue was purified by chromatography on silica gel to afford **2a** (8.3333 g, 83%) (eluent: petroleum ether/ethyl acetate = 500/1 (2 L) to 200/1 (3 L) to 100/1 (1 L) to 2/1 (2.3 L)) as an oil: ^1H NMR (400 MHz, CDCl_3) δ : 11.27 (brs, 1H, COOH), 2.35 (t, J = 7.6 Hz, 2H, CH_2), 1.63 (quint, J = 7.4 Hz, 2H, CH_2), 1.39–1.20 (m, 16H, $8 \times \text{CH}_2$), 0.88 (t, J = 6.8 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, CDCl_3) δ : 180.5, 34.1, 31.9, 29.6, 29.4, 29.3, 29.2, 29.0, 24.7, 22.7, 14.1.

Supporting Information

The supporting information for this article is available on the WWW under <https://doi.org/10.1002/cjoc.201700576>.

Acknowledgement

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